

Stage 2 VLM Wheel-Mask Filter: Micro Report

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Executive Summary

We tested a two-stage wheel-mask pipeline:

1. **Roboflow/YOLO segmentation** proposes candidate wheel masks.
2. **Qwen VLM** receives the original image, a labeled candidate overlay, and candidate metadata, then returns candidate IDs to keep or reject.

The experiment confirms that a VLM filter can reject wet-road wheel reflections while keeping real physical wheels. On the test image, **Qwen3-VL-30B-A3B-Instruct** matched the **Qwen3-VL-235B-A22B-Instruct** quality benchmark and is the better default production candidate for now.

Recommended strategy:

Default: `qwen/qwen3-vl-30b-a3b-instruct`

Fallback: `qwen/qwen3-vl-235b-a22b-instruct`

Trigger fallback when confidence is low or scene geometry is ambiguous.

Test Image

The image contains two real visible wheels and multiple wet-road reflections.



Figure 1: Original car image

Candidate Overlay

Roboflow returned four **wheel** candidates when the pre-filter was changed from **top_n=2** to **no top-N limit**:

- 0: front real wheel
- 1: front wheel reflection
- 2: road reflection / artifact

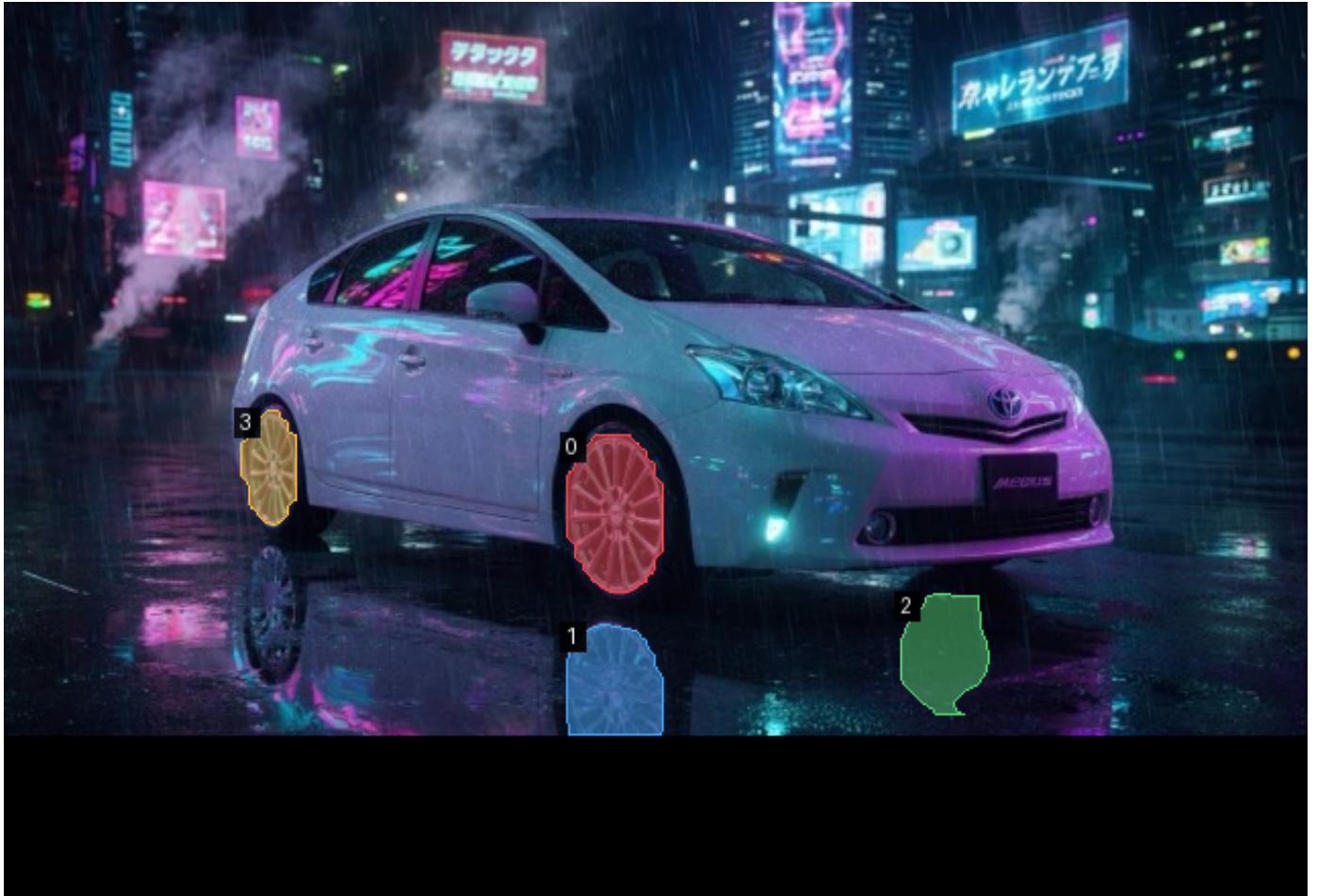


Figure 2: Labeled wheel candidates

- 3: rear real wheel

Key learning: VLM filtering should receive **all reasonable wheel candidates**, not only the largest two. With `top_n=2`, the small rear wheel was excluded before the VLM could evaluate it.

Model Comparison

All models were tested on the same Roboflow candidates with `--top-n 0`.

Model	Keep IDs	Reject IDs	Confidence	Usage	Result
qwen/qwen3-vl-235b-a22b-instruct	[0, 1, 2]	[1, 2]	0.95	1037 in + 72 out	Correct
qwen/qwen3-vl-30b-a3b-instruct	[0, 1, 2]	[1, 2]	0.93	1097 in + 91 out	Correct
qwen/qwen2.5-vl-72b-instruct	[0]	[1, 2, 3]	0.90	1275 in + 64 out	Incorrect: rejected rear wheel

The 30B Qwen3-VL model preserved both real wheels and rejected both reflection candidates. The older Qwen2.5-VL-72B model incorrectly rejected the rear wheel, despite being larger by parameter count.

Final Mask From Qwen3-VL-30B

The final mask contains the two physical wheels only.



Figure 3: Final mask from Qwen3-VL-30B

For comparison, the Qwen2.5-VL-72B run kept only the front wheel:

Notes On The “Missing” Third Reflection

The left-side rear-wheel reflection is visible in the photo, but Roboflow did not return it as a **wheel** candidate. This is not a blocker for production: if Roboflow does not propose a mask, that region cannot leak into the final mask.

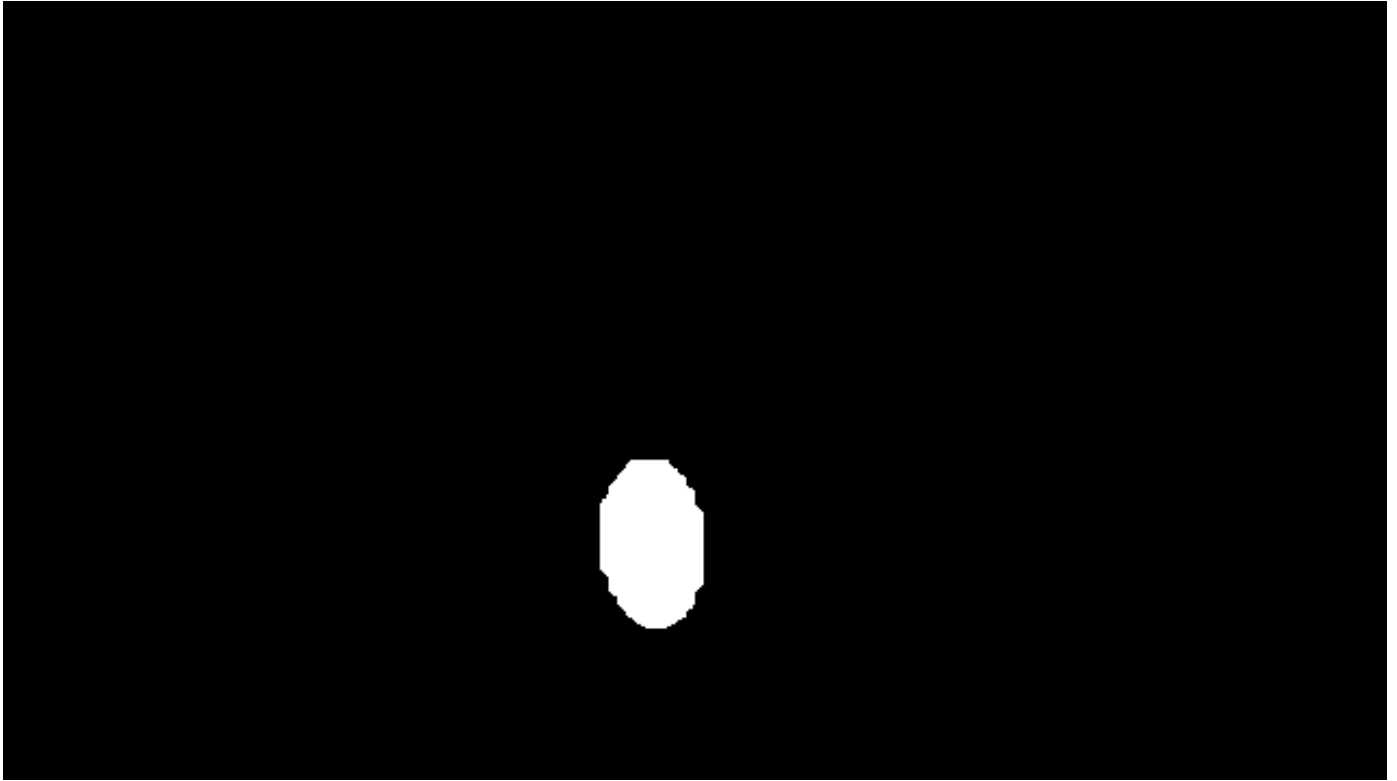


Figure 4: Failure mask from Qwen2.5-VL-72B

For debugging, the candidate overlay now adds bottom padding so candidates near the image border are easier to inspect.

Current Implementation

The probe script is:

`scripts/vlm_mask_filter_probe.py`

Example command:

```
.venv/bin/python scripts/vlm_mask_filter_probe.py \  
  webapp/cover.jpg \  
  tmp/roboflow-benchmark/wheels-tires-body_1/cover.roboflow.json \  
  --top-n 0 \  
  --model qwen/qwen3-vl-30b-a3b-instruct \  
  --output-dir tmp/vlm-mask-filter/qwen3-30b-all
```

Required provider config:

```
VLM_BASE_URL=https://api.zveno.ai/v1  
VLM_API_KEY=...  
VLM_MODEL=qwen/qwen3-vl-30b-a3b-instruct
```

Dataset Recommendation

Current local dataset:

`/Users/nikolai/Documents/Dream Wheel AI/Ivan's Dataset/cars/`

Observed contents:

- 40 image files, about 56 MB total.
- 14 top-level source images, including car photos and rim images.

- 26 generated/result images under RESULTADOS/.

Recommendation:

1. **Do not put the full raw image dataset directly into the main Git repo.** GitHub warns at files over 50 MiB and blocks files over 100 MiB; even smaller image datasets make clones slower over time. Use Git only for metadata, scripts, small curated examples, and benchmark reports.
2. **Use Roboflow for annotation and model-training datasets.** Roboflow is designed for uploading images, videos, and annotations into a project, supports web upload for datasets under 1,000 images, and supports dataset versions with preprocessing and augmentation. This is a better fit for wheel/tire/body masks than plain Git.
3. **Keep a local canonical dataset manifest.** Store a CSV/JSON manifest in Git with image IDs, source path, split, tags, license/consent status, and whether the image is a raw input or generated output. Keep the binary images in Roboflow, local storage, cloud storage, or Git LFS if we need repo-linked storage later.
4. **Separate raw inputs from generated outputs.** The RESULTADOS/ images are useful for product QA, but they should not be mixed into a segmentation training set unless explicitly tagged as generated/synthetic outputs.
5. **Prefer collecting real photos over scraping the internet.** For this task, the hard cases are domain-specific: wet roads, reflections, occluded wheels, dark wheels, side angles, low light, glossy floors, cropped cars, and after-market rims. Real user-like photos will be more valuable than generic internet images. Internet images can help with long-tail diversity, but only if licensing/usage rights are clear.

Near-term target:

50-100 curated real car photos

```
-> tag reflection/lighting/angle conditions
-> run Roboflow candidates
-> run Qwen3-VL-30B filter
-> manually review keep/reject decisions
-> escalate failures to Qwen3-VL-235B
```

References

- Roboflow data upload docs: <https://docs.roboflow.com/datasets/adding-data>
- Roboflow dataset versions: <https://docs.roboflow.com/datasets/dataset-versions>
- Roboflow augmentation guidance: <https://docs.roboflow.com/datasets/dataset-versions/image-augmentation>
- GitHub large file guidance: <https://docs.github.com/en/repositories/working-with-files/managing-large-files/about-large-files-on-github>

Appendix: Negative Examples And Current Risk Status

Dataset Naming

The downloaded Roboflow Universe dataset should be treated as a **Stage 1.5 candidate-segmentation sanity dataset**, not as the final Stage 2 VLM benchmark.

Source dataset: wheeltirebody/wheels-tires-body

Downloaded version: 2

Local path: /private/tmp/dw-roboflow-stage2/wheels-tires-body-v2-coco

Roboflow project copy: the-hishnik/dream-wheels-stage2-wheel-benchmark

Stage label: Stage 1.5 - wheel candidate sanity / negative examples

This dataset is useful for checking whether the Roboflow candidate stage proposes wheel masks on normal car images and avoids firing on images where only body/tire are annotated. It is not sufficient to prove reflection handling.

Stage 1.5 Sanity Benchmark Result

We ran the current Roboflow candidate model on all 125 images:

```
.venv/bin/python scripts/roboflow_benchmark.py \
/private/tmp/dw-roboflow-stage2/wheels-tires-body-v2-coco \
--model-id wheels-tires-body/1 \
--classes wheel \
```

```
--top-n 0 \
--output-dir tmp/stage15-wheel-candidate-benchmark
```

Summary against the COCO annotations at the image level:

Metric	Result
Total images	125
Images with ground-truth wheel	82
Images without ground-truth wheel	43
True positives	79
True negatives	43
False positives	0
False negatives	3
Image-level accuracy	97.6%
Wheel-image detection rate	96.3%
No-wheel negative pass rate	100.0%

Interpretation:

- The current Roboflow model is a good candidate generator on this dataset.
- It did not produce wheel candidates on the 43 no-wheel negative examples.
- It missed wheel candidates on 3 images that had wheel annotations.
- This is an image-level sanity check, not a pixel-level mask-IoU evaluation.

Negative Examples: Current Coverage

Current negative coverage is uneven:

Negative type	Current status	Notes
Body-only / no-wheel car views	Covered by Stage 1.5 dataset	43 no-wheel images passed with no wheel false positives.
Tire/body vs wheel confusion	Partially covered	Dataset includes body , tire , and wheel ; useful for candidate sanity.
Wet-road reflections	Weak coverage	The synthetic/stylized cover.jpg is currently the main validated case.
Glossy floor / showroom reflections	Not enough coverage	Needs a dedicated hard-case set.
Shadows shaped like wheels	Not enough coverage	Needs real user-like photos.
Background vehicles / background wheels	Partially covered	Some dataset images include multiple cars, but not a controlled benchmark.
Cropped or partial wheels	Partially covered	Present in dataset, but no separate metric yet.
Night / low-light / neon glare	Weak coverage	cover.jpg covers one stylized example only.
Generated/synthetic outputs	Kept separate	Ivan's RESULTADOS/ should not be mixed into raw training/benchmark data unless explicitly tagged.

Pipeline Risks

The main risk is **candidate recall**. If Roboflow does not propose a physical wheel mask, the VLM cannot recover it. The VLM only selects from existing candidate IDs; it does not draw masks.

Other current risks:

- **Top-N truncation risk:** using `top_n=2` dropped the small rear wheel in the **cover** experiment. Stage 2 should pass all reasonable wheel candidates after class/min-area filtering.
- **Reflection hard cases are under-tested:** the current success on wet-road reflections is promising but based on one strong example.

- **Pixel-level quality is not yet measured:** the Stage 1.5 benchmark checks whether at least one wheel candidate exists, not mask IoU or wheel completeness.
- **Dataset domain mismatch:** Universe images are useful but skew toward internet/tuning/showcase photos. Telegram user photos may be blurrier, darker, more compressed, and less centered.
- **Class semantics mismatch:** some datasets annotate **tire**, some annotate **wheel**, some annotate rim-like regions. We need a consistent definition for the final inpainting mask.
- **Provider/model drift:** OpenAI-compatible providers can change routing, model availability, pricing, or output formatting. The script should keep JSON validation and fallback behavior strict.
- **Privacy and licensing:** user photos should not be uploaded to public datasets; internet images need clear usage rights.

Current Gaps Before MVP Stage 2 Confidence

Before calling Stage 2 production-ready, we still need:

1. A dedicated **Stage 2 hard-negative benchmark** with 30-50 images containing wet roads, reflections, glossy floors, shadows, night scenes, and background wheels.
2. A batch VLM evaluation script that runs **Qwen3-VL-30B** across benchmark images and stores keep/reject decisions in a reviewable CSV/JSON.
3. Manual review labels for candidate decisions:
 - physical wheel,
 - reflection,
 - shadow,
 - background wheel,
 - body/tire artifact,
 - missed physical wheel.
4. Pixel-level mask quality checks for final masks, at least on a small manually reviewed subset.
5. Escalation logic:

Qwen3-VL-30B default

- > if confidence is low, no candidates are kept, or scene has reflection-like geometry
- > retry with Qwen3-VL-235B

Practical Next Step

Use the Stage 1.5 dataset to validate candidate generation, but build a separate Stage 2 hard-negative set. The minimum useful next benchmark is:

30-50 hard images

- 10 wet-road reflections
- 10 glossy floor / showroom reflections
- 10 low-light or night images
- 5-10 cropped/occluded wheels
- 5-10 background-wheel or multi-car scenes

The Stage 1.5 result is encouraging, but it does not replace this hard-negative benchmark.